

Engine Only Operates in Droop Mode

Symptoms

- The engine will **only** operate in droop mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine droop symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Droop Adjust Potentiometer Supply and Signal Wire for Short Circuit	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Droop Adjust Potentiometer Supply and Signal Wires	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Droop Adjust Potentiometer Supply and Signal Wire for Short Circuit

- Conditions :**
- Open the customer interface box
 - Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer supply and signal wires. • Disconnect the droop adjust 5 volt supply (sensor supply 1) wire and the droop adjust potentiometer signal wire from the X4 connector. • Place one test lead on the droop adjust potentiometer 5 volt supply (sensor supply 1) pin in connector C2. • Place the other test lead on droop adjust potentiometer signal pin in connector C2.	Less than 10 ohms resistance? YES Repair : Replace the wire(s). Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete.
	Less than 10 ohms resistance? NO	2A

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Droop Adjust Potentiometer Supply and Signal Wires

Conditions :

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step

Check droop adjust potentiometer supply and signal wires. - Place one test lead in the droop adjust potentiometer 5 volt supply (sensor supply 1) pin of the C2 connector. - Place the other test lead in the droop adjust potentiometer signal pin of the C2 connector.	Less than 10 ohms resistance? YES Repair : Refer to Section TF in the Troubleshooting and Repair Manual, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html), or refer to the OEM service manual for potentiometer repair instructions.	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

